

# An unmeasured observable in two-channel Bell experiments

Proposal for a settings-torus scan of the fourfold coincidence sum

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## The observable

In a two-channel polarization Bell experiment with analyzer settings  $(a, b)$ , the fourfold coincidence sum

$$R_4(a, b) = R_{++} + R_{+-} + R_{-+} + R_{--}$$

is predicted flat by quantum mechanics under fair sampling, and has long been recognised as the natural fair-sampling diagnostic. To my knowledge its angular dependence has never been bounded below the percent level: the landmark two-channel experiment reported the sum only as “typically  $\sim 80 \text{ s}^{-1}$ ”, the percent-level setting-dependent anomalies in the Innsbruck raw data were attributed to pair-identification procedures rather than bounded, and the modern high-efficiency tests adopted the Eberhard/CH form — a rigorous sidestep that leaves the observable unmeasured. Full record: arXiv:2608.18886, §9.

Two results in arXiv:2608.18886 bear on this sum. First, within the detection-model class the paper treats (deterministic outcomes, a common detection profile, clicks independent given the hidden phase), no model that reproduces the singlet correlation exactly can also have an angle-independent coincidence rate (Proposition 1); some angular modulation is forced. Second, for the exact detection profile  $D(m) = \sqrt{m} h(m)$  solved in §5 of the paper, the modulation is a  $\cos 4(a - b)$  fringe of peak-to-peak relative amplitude 12.4%, leading harmonic exceeding its first overtone  $\approx 6:1$ . Hybrid models paying partly through measurement dependence show proportionally reduced amplitude; the trade-off is quantified by linear programming along that family ( $\delta_f < 1\%$  forces  $\geq 68\%$  of Hall’s measurement-dependence floor;  $\delta_f < 0.5\%$  forces  $\geq 80\%$ ).

**Why no existing dataset bounds it.** The fringe takes identical values at the two CHSH relative angles,  $22.5^\circ$  and  $67.5^\circ$ , harmonic by harmonic:  $\cos(2k \cdot 45^\circ) = \cos(2k \cdot 135^\circ)$  for every  $k$ , so  $R_4(22.5^\circ) = R_4(67.5^\circ)$  exactly, term by term (arXiv:2608.18886, Theorem 2). Four-point CHSH experiments are therefore mathematically blind to the observable, and the Eberhard/CH-form loophole-free tests did not record  $R_4$  at all. Only a full settings scan sees it.

## The measurement

Scan the full settings torus  $(a, b)$  — e.g.  $24 \times 24$  points — and Fourier-analyse  $\log R_4(a, b)$ . Any local apparatus response factorizes as  $A(a)B(b)$  and after the logarithm is additive: it occupies only the Fourier axes, at any amplitude, harmonic content, or drift phase. The physical signal  $\cos 4(a - b)$  is irreducibly joint and occupies the anti-diagonal pixels  $(k, -k)$ . The estimator is therefore self-vetoing against factorizable systematics. Because the fringe has known (zero) phase, the estimator is the signed projection onto the  $(2, -2)$  pixel; a magnitude estimator would be biased positive under noise and is not used. The signed projection assumes the relative zero of the two analyzers is calibrated: a relative offset  $\delta$  rotates signal into the quadrature and costs a factor  $\cos 4\delta$  ( $0.3\%$  at  $\delta = 1^\circ$ ). The quadrature (imaginary) component of the same pixel must be null-consistent in every dataset, signal or none, and therefore serves as a built-in systematic check.

In simulation with 3% apparatus ripple and Poisson noise at  $10^6$  counts per point, a null input is

recovered as  $(0.00 \pm 0.01)\%$  and the estimator is linear from 0.2% through the full 12.4%. Poisson-limited  $5\sigma$  sensitivity at  $\delta_f = 0.1\%$ , measured against the null distribution, is reached at  $\sim 4 \times 10^5$  counts per point ( $\sim 2.3 \times 10^8$  pairs); the paper quotes the deliberately conservative figure of  $\sim 7$  hours at  $5 \times 10^4$  pairs/s, or  $\sim 20$  minutes at  $10^6$  pairs/s, and that is the figure to plan against. Controls: randomized visiting order and heralded-singles normalization against source drift; interleaved reference points against slow systematics; linear-regime detector rates; quadrature null check at the signal pixel.

**Requirements.** A standard polarization-entangled source, independently rotatable two-channel analyzers, and coincidence rates  $\gtrsim 5 \times 10^4$  pairs/s. No loophole-free rigour is needed: the observable is a rate map, not an inequality violation.

## Outcomes

Every branch is publishable:

1. **Flatness at 0.1%** — to my knowledge the first sub-percent bound on a diagnostic left un-measured for four decades. Within the faithful detection family of arXiv:2608.18886, it eliminates the detection share and forces  $\gtrsim 95\%$  of Hall’s measurement-dependence floor onto the measurement-dependence channel.
2. **A fringe of the predicted shape** — a flat-rate prediction of quantum mechanics fails.
3. **A fringe of a different shape** — the detection sector is excluded as constructed, and a joint  $(a, b)$  dependence that no factorizable apparatus model can produce is characterised.

## Materials

Paper: arXiv:2608.18886 (trade-off surface, mechanism, Theorem 2, protocol §10).

Scripts, including the sensitivity simulation above (`protocol_sensitivity.py`): `Exact_Price_Source_Material` folder of the `moire-phase-space-sampler` repository (GitHub, user `RandomInternetPreson`).

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